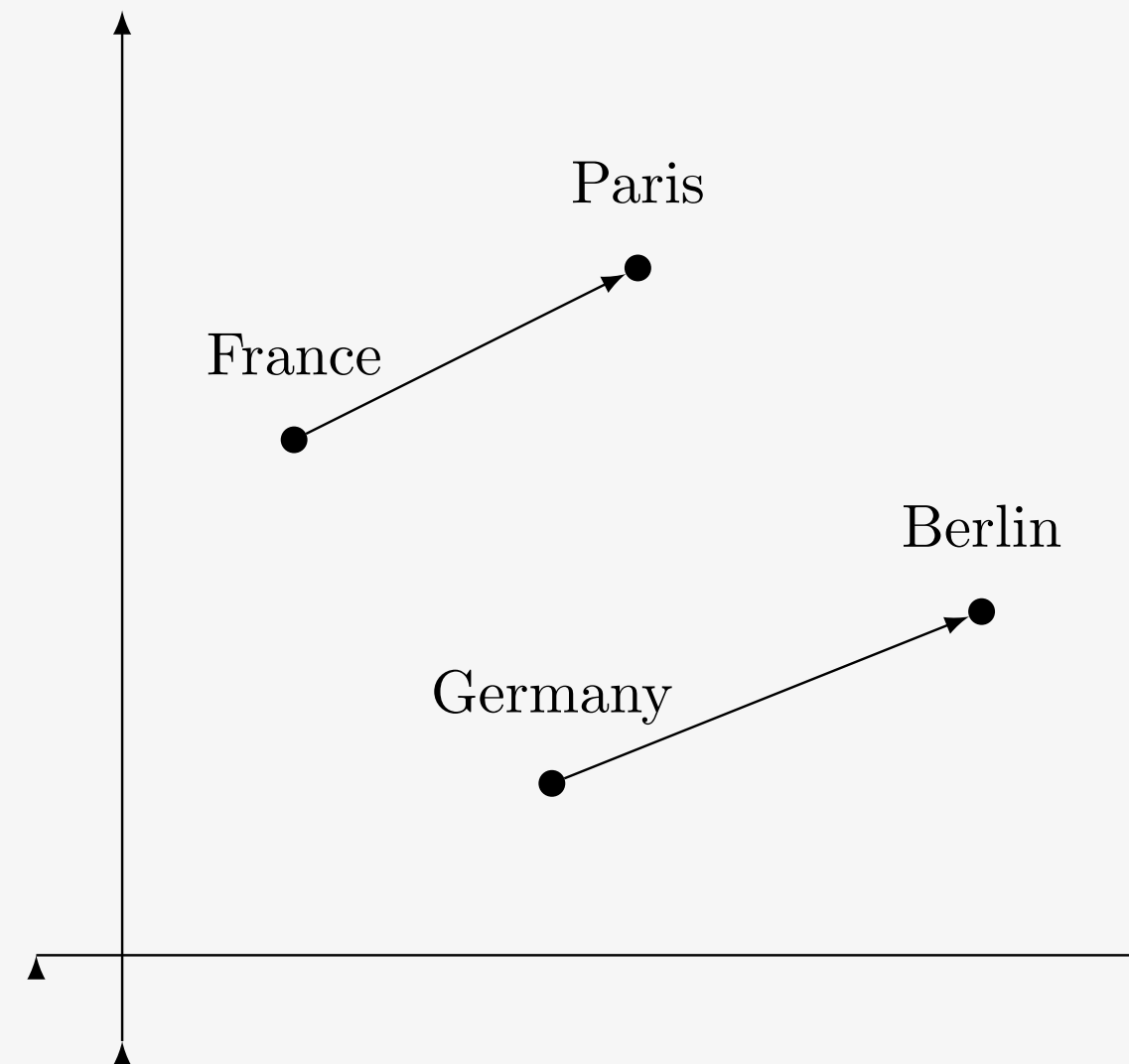


WORD2VEC

Mahdi Meskin

EFFICIENT ESTIMATION OF WORD REPRESENTATIONS IN VECTOR SPACE

how can a computer learn that France, Paris, Germany and Berlin are related?



ONE-HOT VECTORS

- Unique representations
- No meaning or relations

$$\textit{France} = [1, 0, 0, 0]$$

$$\textit{Paris} = [0, 1, 0, 0]$$

$$\textit{Germany} = [0, 0, 1, 0]$$

$$\textit{Berlin} = [0, 0, 0, 1]$$

PAPER GOALS

1. Words with similar meanings sit close together in continuous space
2. Replaced sparse million-item lists with compact 300-dimension feature vectors.

THE ARCHITECTURES

1

CBOW

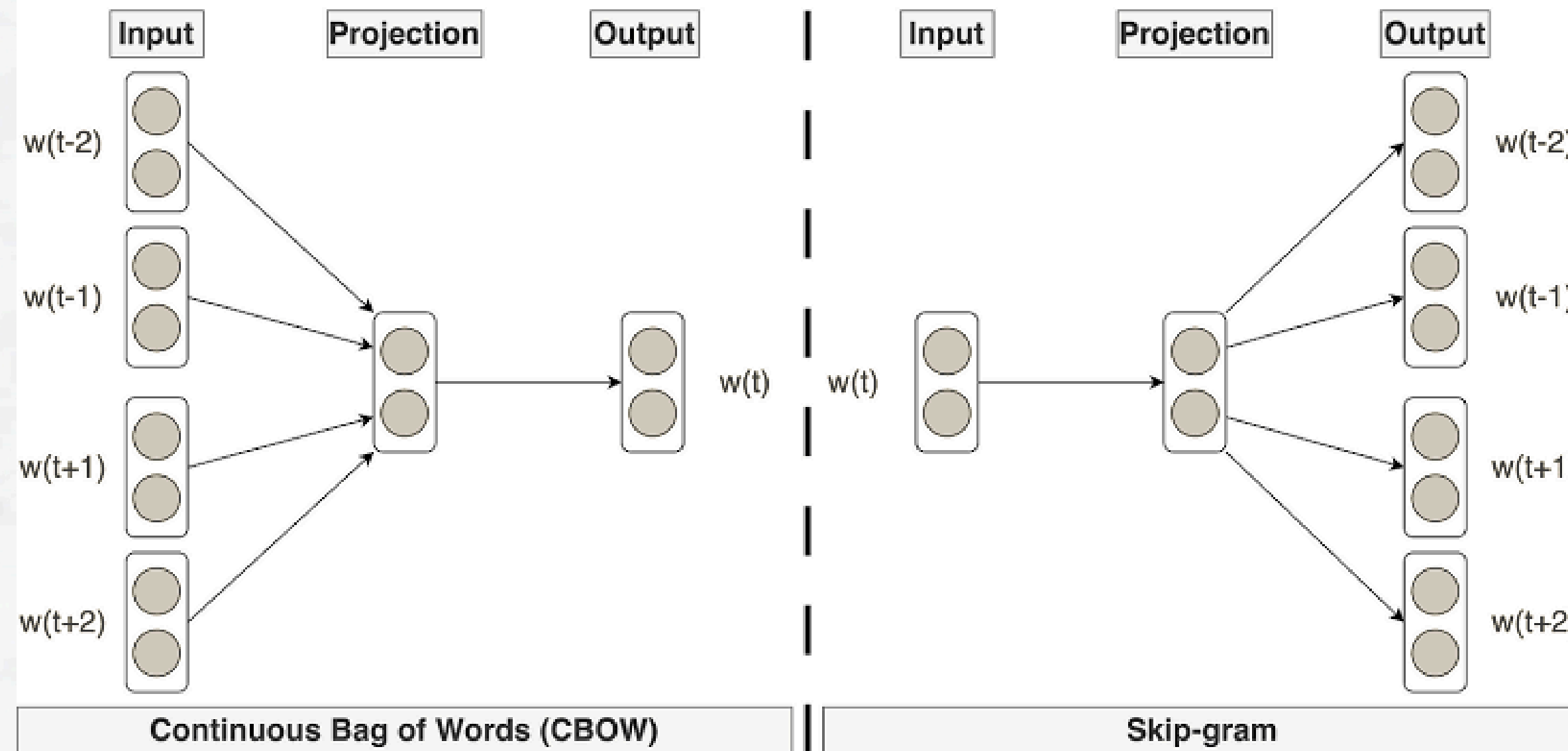
Continuous Bag of Words: Predicts center word using surrounding context.

2

Skip-Gram

Continuous Skip Gram: Predicts surrounding context using center word.

Word2Vec



CBOW

1. Collect context words
2. Averages context vectors together into one combined layer
3. Predicts missing center target word

SKIP-GRAM

1. Takes single target center word
2. Projects target vector across weight matrix.
3. Predicts surrounding words

WORD RELATIONSHIPS

1

Sytactic

adjective to verb rapid → rapidly

past tense walk → walked

2

Semanticc

Man → Woman Brother → Sister

France → Paris Germany → Berlin



Q & A SESSION



THANK
YOU